



WeNode — Technical System Spec

WeNode is a sovereign distributed compute network that allows anyone to contribute compute capacity through either a browser-based WASM node or a native node operator. The system accepts jobs, splits them into shards, distributes them across available nodes, and aggregates results. All operator payouts are settled in USD via Stripe, making the network accessible and compliant from day one.

The platform is built on three fully implemented layers:

1. COMPUTE LAYER (EXECUTION ENGINE)

This is the operational core of WeNode. It runs workloads, distributes shards, and returns results.

1.1 Components

- a. WASM Nodes
 - Executed in the browser
 - WebAssembly sandbox
 - Poll for work
 - Execute shards
 - Return results
 - Send heartbeats with telemetry
- b. Native Node Operator
 - Go binary
 - Runs on user machines or servers
 - Polls for work
 - Executes shards
 - Returns results
 - Sends heartbeats with hardware metadata
- c. Distributed Engine
 - Accepts jobs
 - Splits jobs into shards
 - Selects nodes
 - Assigns shards
 - Handles timeouts
 - Handles retries

- Aggregates results
- d. Heartbeat System
 - Nodes send periodic heartbeats
 - Backend updates:
 - uptime
 - performance
 - reputation inputs
 - identity consistency
- e. Payout Engine (Stripe)
 - Operators are paid in USD
 - Stripe handles:
 - payouts
 - settlement
 - compliance
 - reporting

1.2 How it works

1. A job is submitted.
2. Distributed engine splits it into shards.
3. Nodes poll for work.
4. Shards are assigned based on:
 - availability
 - reputation
 - identity trust
5. Nodes execute shards and return results.
6. Results are aggregated on clients source machine
7. Operators are paid in fiat.

All of this is implemented and verifiable.

2. **ECONOMIC SAFETY LAYER (INTERNAL MECHANISMS)**

This layer protects the network from abuse and ensures reliability.
It is internal, not user-facing, and not a financial system.

It contains three subsystems:

- Internal accounting

- Collateral locking
- Reputation scoring

All three are implemented.

2.1 INTERNAL ACCOUNTING

This is an internal ledger used by the backend to track operator activity.

Components

- Internal balance ("WEX")
- Ledger entries for:
 - rewards
 - penalties
 - slashes
 - locked collateral
- Spendable vs locked balance
- Transaction history

How it works

- Completing work → internal credit
- Abandoning work → internal debit
- Receiving a shard → collateral locked
- Completing a shard → collateral unlocked

This system is fully implemented and visible in:

- `nodld/internal/account/tokens.go`
- `nodld/internal/account/persist.go`
- Command dashboard

2.2 COLLATERAL LOCKING

A protective mechanism that ensures nodes behave reliably.

Components

- Minimum internal collateral requirement
- Collateral locked per shard
- Slashing logic
- Stake ledger
- Stake dashboard

How it works

1. Node requests work.
2. Backend locks a small amount of internal collateral.
3. If shard completes → collateral returned.
4. If shard abandoned → collateral slashed.

This is implemented in:

- `staking.go`
- `reputation.go`
- `distributed.go`

2.3 REPUTATION ENGINE

A deterministic trust score used to improve scheduling and reliability.

Components

- Reliability
- Uptime
- Success rate
- Abandonment rate
- Slashing count
- Compute performance
- Longevity
- Stake factor
- Identity trust factor
- Reputation ledger
- Reputation dashboard

How it works

- Heartbeats update metrics
- Shard results update metrics
- Daily cron recalculates all reputations
- High-rep nodes get more work
- Low-rep nodes get fewer shards

This is implemented in:

- `reputation.go`
- `store.go`

- Command dashboard

3. IDENTITY & INTEGRITY LAYER

This layer ensures both humans and devices are uniquely and consistently identified.

It contains:

- WUID (human identity)
- Device fingerprinting

Both are implemented.

3.1 WUID — HUMAN IDENTITY

WUID (Wenode Universal ID) is the permanent identity of a human in the Wenode ecosystem.

Components

- WUID registry
- Soul assignment
- Governance identity
- Account ownership

How it works

- Each human has one WUID
- WUID is used for:
 - governance
 - Souls
 - account control
 - operator ownership

This is part of your existing governance system.

3.2 DEVICE FINGERPRINTING

Ensures each node (browser or native) is uniquely identifiable.

Components

a. Native Nodes

- hardwareHash computed from:
 - CPU model

- core count
- flags
- RAM
- disk model
- machine ID
- OS version

b. WASM Nodes

- browserFingerprint computed from:
 - canvas fingerprint
 - WebGL vendor/renderer
 - audio fingerprint
 - user agent
 - timezone
 - screen resolution
 - fonts hash

c. Backend Identity System

- deviceClass
- trustLevel
- identity ledger
- Sybil detection
- linked node detection

How it works

- Nodes send fingerprints on register + heartbeat
- Backend evaluates:
 - consistency
 - stability
 - duplicates
 - anomalies
- trustLevel influences:
 - scheduling
 - collateral
 - reputation

This is implemented in:

- `identity.go`
- `model.go`
- `store.go`
- Command identity dashboard

4. AI Observation Layer (Passive Mode Only)

Wenode includes an AI model that is integrated into the system in a **read-only, observation-only mode**. The model is active, running, and capable of learning, but it does not influence scheduling, economics, governance, or any operational parameters yet. These additional functions will be activated when the system is hosted on better hardware.

The system already emits all the signals required for future optimisation, including:

- node heartbeats
- shard execution metadata
- reputation deltas
- identity trust changes
- collateral lock/unlock events
- slashing events
- scheduling decisions
- job metadata (never job data)

These signals allow an AI model to observe patterns in reliability, performance, identity consistency, operator behavior, and workload characteristics. The AI layer can ingest this telemetry and build internal representations, embeddings, and insights, but **cannot modify system behavior**.

The architecture is fully compatible with the incoming AI optimisation layer. The AI model does not tune parameters, propose changes, or influence the network. It operates purely as a passive learner whilst in training mode.

This ensures complete safety, verifiability, and zero operational risk while laying the foundation for the near term future AI-driven improvements.

FINAL VERIFIED SUMMARY

Wnode today consists of:

1. Compute Layer

- WASM nodes
- Native nodes
- Distributed engine
- Heartbeats
- USD payouts via Stripe

2. Economic Safety Layer

- Internal accounting
- Collateral locking
- Slashing
- Reputation scoring

3. Identity & Integrity Layer

- WUID (human identity)
- Device fingerprinting
- Identity trust
- Sybil detection

4. No AI control layer exists yet

- The system is architected to support this in near term iterations